

# Three Beats Eight: Coordination, Not Fleet Size, Wins Multi-Agent Disaster Rescue

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**Abstract**—When several autonomous rescue units serve one disaster area, the cheapest way to lose time is for two of them to drive to the same victim. Coordination protocols prevent this, but they differ sharply in communication cost, and post-disaster networks are lossy. We ask which part of coordination buys the gain, and how the answer changes with team size, victim arrival dynamics, and message loss. We extend RESQ-MAS, an open multi-agent rescue simulator, into a testbed of six allocation protocols: independent selection, claim broadcasting, one-round and iterative propose–resolve negotiation, the consensus-based bundle algorithm (CBBA), and a centralized Hungarian matcher. Across 43,350 paired simulations, one-round negotiation cuts mission time by 29.0% for two agents and by up to 66% for eight (Holm-corrected Wilcoxon  $p < 10^{-4}$ ). Uncoordinated teams saturate: eight independent agents are only  $1.7\text{--}2.5\times$  faster than one, and three negotiating agents beat eight independent ones on every workload. The value of negotiation over claim broadcasting is set by contention, rising from 4% of the gain at 15 victims per agent to 61% at 3, and it dominates (55–62%) once victims arrive over time, where coordination cuts mean victim wait by 72–79%. Iterative negotiation and CBBA both match the centralized optimum, but CBBA sends 5–56 $\times$  more data. That redundancy buys robustness to independent message loss (80% of the gain retained at 50% loss, against 56%), yet little of it survives bursty outages. All raw runs are released.

**Index Terms**—Multi-agent systems, multi-robot task allocation, disaster response, search and rescue, decentralized coordination, auction algorithms, Hungarian algorithm, price of anarchy, lossy communication, agent-based simulation.

## I. INTRODUCTION

After an earthquake or flood, emergency operations centers dispatch ambulances, ground robots, and drones to many casualties at once, often over degraded radio links and with little time for central planning [1], [2]. A recurring and expensive failure mode in such operations is *duplicated effort*: two units independently decide that the same critical victim is the most urgent, both travel there, and one arrives to find nothing to do while other victims keep waiting. Multi-agent coordination research offers many remedies, from market-based auctions [3] to consensus-based and distributed optimization methods [4], [5] and centralized optimal assignment [6]. What a practitioner designing a dispatch system actually needs to know, however, is more specific: *which part* of coordination produces the benefit, *how much communication* it costs, and *how it behaves* when messages are lost.

This paper answers these questions with a controlled, paired simulation study. Our starting point is RESQ-MAS, an open-

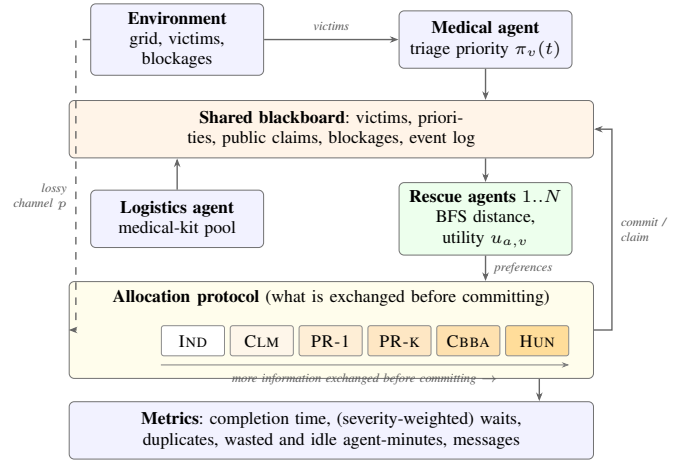


Fig. 1. RESQ-MAS testbed. The world is generated deterministically from a seed, so every protocol runs on an identical world (paired design). The medical agent publishes priorities to a shared blackboard; rescue agents score reachable victims and hand their preferences to one of six allocation protocols that differ only in what is communicated before an agent commits. Routing, replanning, pickup, and delivery are shared code, so any metric difference is attributable to the protocol alone.

source disaster-rescue simulator in which a medical triage agent, a logistics agent, and a team of rescue agents interact through a shared blackboard. Its original release compared two policies on 20 hand-picked runs. We extend it into a protocol testbed (Fig. 1) with six allocation protocols, including the widely used consensus-based bundle algorithm (CBBA) [4], support for arbitrary team sizes, victims that arrive over time, independent and bursty message-loss channels, and instrumentation for wasted effort and message volume. We evaluate it on 43,350 simulations over randomly generated worlds.

## A. Contributions

- A **communication ladder** of six task-allocation protocols implemented on one shared agent codebase, so that the *only* difference between conditions is what agents exchange before committing: nothing, public claims, one negotiation round, iterative negotiation, CBBA bundle consensus, or full utility reports to a centralized optimal matcher.
- A **decomposition of the coordination gain** into a claim-broadcasting component and a same-tick negotiation component, showing that the share due to negotiation is

governed by contention (4% to 61%) and dominates once victims arrive over time.

- Evidence that **uncoordinated teams saturate**: adding agents without coordination yields at most  $2.5\times$  speedup at eight agents, the empirical price of anarchy grows roughly linearly with team size, and *three coordinated agents outperform eight independent ones*.
- Evidence that **decentralized iterative negotiation matches a centralized matcher and CBBA** in all 15 tested configurations with  $3\text{--}11\times$  less data than the matcher and  $5\text{--}56\times$  less than CBBA, and that CBBA's extra traffic buys robustness to independent loss but little against bursty outages.
- A reproducible, open testbed with fixed seeds, paired statistics, and all raw runs released.

## II. RELATED WORK

### A. Multi-Robot Task Allocation

Recent surveys [6]–[8] organize multi-robot task allocation (MRTA) by whether a robot executes one or several tasks, whether a task needs one or several robots, and whether assignment is instantaneous or planned over time; Nunes *et al.* [9] extend this taxonomy to temporal and ordering constraints. Our setting, in which each agent serves one victim at a time and victims are known up front, is the single-task, single-robot, instantaneous-assignment class, repeated as agents become free. Each decision epoch is a linear assignment problem, solvable optimally by the Hungarian method, which Chopra *et al.* [5] showed can also be executed in a fully distributed way over a peer-to-peer network. Quinton *et al.* [3] review market-based MRTA, from Contract-Net-style single-item auctions to consensus-based bundle algorithms, and report that greedy auctions routinely come close to optimal allocations at a fraction of the communication cost. Rizk *et al.* [10] survey cooperative heterogeneous teams, including behaviour-based schemes that coordinate implicitly by broadcasting current activity, which corresponds to the claim-broadcasting rung of our ladder.

### B. Coordination for Disaster Rescue

Queralta *et al.* [1] and Drew [2] survey multi-robot search-and-rescue systems and both single out coordination under unreliable communication as an open operational problem. Ghassemi and Chowdhury [11] formulate disaster-response MRTA with task deadlines and robot range and payload constraints and solve it with a decentralized bigraph-matching approach, showing that the allocation layer materially changes how many victims are served in time.

### C. Coordination under Limited Communication

Buckman *et al.* [4] extend the consensus-based bundle algorithm with partial replanning, trading convergence time for coordination quality when new tasks appear online. Otte *et al.* [12] compared auction variants under message loss and found that performance degrades gracefully as loss increases. Gielis *et al.* [13] review communication in multi-robot systems

TABLE I  
POSITIONING RELATIVE TO REVIEWED WORK

| Study                                     | Focus                                                  | Gap addressed here                                                                                                         |
|-------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Quinton [3],<br>Chakraa [6]<br>Chopra [5] | MRTA surveys<br>(2023)<br>Distributed<br>Hungarian     | Qualitative comparison; no controlled ladder of protocols<br>Optimality with full exchange; cost of less talk not measured |
| Ghassemi [11]<br>Otte [12]                | Disaster MRTA<br>with deadlines<br>Auctions under loss | One method; no independent or claim-only baselines<br>No decomposition of the gain or team-size scaling                    |
| Lott [14]                                 | Allocators under<br>degraded links<br>(2026)           | Consensus/auction family only; no rescue priorities or claim-only rung                                                     |
| <b>This paper</b>                         | Five-rung ladder,<br>28.3k paired runs                 | Decomposition, scaling law, optimality gap, loss robustness                                                                |

and argue that its cost and reliability should be treated as first-class design variables rather than assumed ideal. Most recently, Lott and Honary [14] benchmarked six decentralized allocators (CBAA, ACBBA, and others) on paired instances under Bernoulli, Gilbert–Elliott, and fading channels, comparing travel cost, reliability, and computation. On the game-theoretic side, Paccagnan *et al.* [15] characterize the price of anarchy, the ratio between the cost of uncoordinated equilibria and the social optimum, for distributed resource allocation, which gives a natural lens for measuring what coordination is worth.

### D. Research Gap

Existing studies typically propose one protocol, or benchmark several allocators of the same family, so the *marginal* value of each additional kind of communication (none, public claims, negotiation, full centralization) is rarely isolated on one codebase. The scaling of that value with team size, the role of contention, and the cost of centralization are usually reported, if at all, for a single configuration. Table I positions this paper.

## III. SYSTEM MODEL

### A. Environment

The world is a  $W \times W$  4-connected grid ( $W = 15$  or  $20$ ) in which about 10% of cells are buildings, one cell is a hospital, and two corner cells are depots from which rescue agents start.  $M$  victims are placed on random road cells with severities drawn from {critical, high, moderate, low} with probabilities {0.2, 0.3, 0.3, 0.2}. A fraction  $b$  of free road cells is scheduled to become blocked at a random time in the first 120 simulated minutes and to reopen 10–75 minutes later; blockages are published to all agents immediately and trigger BFS replanning. Victims are either all present at  $t = 0$  (static) or appear at uniformly random times in  $[0, A]$  for an arrival window  $A$  (dynamic); a victim is unknown to every agent, and accrues no waiting time, until it appears. Arrival times use a separate random stream, so static and dynamic runs share grid, positions, severities, and blockages. One tick represents one simulated minute. A victim counts as rescued

when delivered to the hospital; each rescue consumes one medical kit, allocated by the logistics agent, which refuses a second kit for a victim that already holds one. An agent whose target has been rescued by someone else, or whose kit request stays denied for 40 ticks, abandons that target.

### B. Priorities and Utility

The medical agent republishes each waiting victim's priority as

$$\pi_v(t) = s_v + \lambda w_v(t), \quad (1)$$

where  $s_v \in \{4, 3, 2, 1\}$  is the severity score,  $w_v(t)$  the waiting time so far, and  $\lambda = 0.1$ , so severity dominates but long-waiting victims eventually rise. An idle agent  $a$  scores every reachable candidate victim  $v$  by

$$u_{a,v}(t) = \pi_v(t) - \beta d_a(v), \quad (2)$$

with  $d_a(v)$  the BFS shortest-path length on the currently known grid and  $\beta = 0.1$ . All protocols use the same utility; they differ only in how preferences are turned into commitments.

### C. The Communication Ladder

**IND (independent).** Each idle agent commits to its own  $\arg \max_v u_{a,v}$  over all unrescued victims. Nothing is communicated, so agents may converge on the same victim or chase a victim already being carried.

**CLM (claim broadcasting).** After committing, an agent broadcasts a claim; other agents exclude claimed victims from their candidate sets. Choices made in the same tick are not negotiated, so simultaneous duplicates remain possible. This mirrors implicit coordination by activity broadcasting in behaviour-based teams [10].

**PR-1 (one-round propose–resolve).** Idle agents first broadcast a proposal  $(v, u_{a,v})$ . Proposals for the same victim are resolved deterministically: highest utility wins, ties go to shorter distance, then agent identifier. Winners commit and broadcast claims; losers stay idle for the rest of the tick and propose again at the next tick. The resolution rule is a pure function of public messages, so every agent can compute it locally; no privileged coordinator is needed. This is the original RESQ-MAS coordination mechanism.

**PR-K (iterative propose–resolve).** As PR-1, but losers immediately re-propose their best victim not yet won in the current tick, repeating until every idle agent holds a target or no candidate remains (Algorithm 1). This is a sequential single-item auction [3] executed within one tick.

**CBBA (consensus-based bundle algorithm).** The standard decentralized market baseline [3], [4], re-run in every tick in which some agent is idle. Every agent, including busy ones, greedily builds a bundle of up to  $L = 3$  future victims, scoring a victim at bundle position  $k$  as  $\pi_v - \beta \tau_k$ , where the pickup time  $\tau_k$  chains through hospital drop-offs and, for a busy agent, starts when and where it next becomes free. Agents then broadcast their bundles and each keeps the most recent bundle heard from every other agent; the highest bid wins

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### Algorithm 1 PR-K: one tick of iterative propose–resolve

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1:  $\mathcal{A} \leftarrow$  idle agents;  $\mathcal{T} \leftarrow$  victims won this tick
2: each  $a \in \mathcal{A}$  proposes  $v_a = \arg \max_{v \notin \text{claims} \cup \mathcal{T}} u_{a,v}$ 
3: while  $\mathcal{A} \neq \emptyset$  and some proposal exists do
4:   for each victim  $v$  with proposals do
5:     winner  $\leftarrow \arg \max(u_{a,v}, -d_a(v), -\text{id}_a)$ 
6:     winner commits, broadcasts claim;  $\mathcal{T} \leftarrow \mathcal{T} \cup \{v\}$ 
7:   end for
8:    $\mathcal{A} \leftarrow$  losers; each re-proposes over victims  $\notin \mathcal{T}$ 
9: end while

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each victim, and an agent outbid on a bundle item releases it and every later item. Rounds repeat until no bundle changes, and idle agents commit to the head of their bundle. A lost broadcast leaves receivers with stale bids.

**HUN (centralized reference).** Each idle agent reports its full utility row; a coordinator solves  $\max \sum_{a,v} u_{a,v} x_{a,v}$  over one-to-one assignments with the Hungarian method as implemented in SciPy [16] and returns the matching. HUN is optimal *per decision epoch*, not over the whole mission, because it re-solves only for currently idle agents and never pre-empts commitments.

### D. Lossy Broadcast Model

Under the *independent* model, every broadcast (proposal, bundle, or claim) is lost with probability  $p$  for all receivers, a broadcast-erasure channel. Under the *bursty* model, each agent's uplink is a Gilbert–Elliott channel stepped once per tick: in the Bad state all of that agent's broadcasts are lost, in the Good state none are, and the transition probabilities  $P(B \rightarrow G) = 1/L_b$  and  $P(G \rightarrow B) = p/(L_b(1 - p))$  give long-run loss rate  $p$  with outages lasting  $L_b$  ticks on average. A sender whose proposal is lost hears no objection and commits unilaterally; a lost claim leaves the victim visible as a candidate to others until its delivery to the hospital, which we assume is always observed. Loss draws use a random stream separate from world generation, so lossy and loss-free runs share the same world. At  $p = 1$  every message-based protocol reduces to IND.

### E. Metrics

Per run we record the completion time  $T$  (minute the last victim is delivered), the mean waiting time of rescued victims, a severity-weighted waiting time  $\bar{w}_s = \sum_v s_v w_v / \sum_v s_v$ , the mean wait of critical victims, duplicate commitments (distinct victims to which two agents were simultaneously committed), wasted agent-minutes (time spent committed to targets later abandoned), idle agent-minutes while victims remained, and communication as broadcast count and payload (scalar values transmitted). From these we derive the speedup  $S(N) = T(1)/T(N)$ , the empirical price of anarchy  $\text{PoA} = T_{\text{IND}}/T_{\text{HUN}}$ , the optimality gap of protocol  $X$ ,  $(T_X - T_{\text{HUN}})/T_{\text{HUN}}$ , and the *negotiation share* of the coordination gain,

$$\eta = \frac{T_{\text{CLM}} - T_{\text{PR-1}}}{T_{\text{IND}} - T_{\text{PR-1}}}, \quad (3)$$

the fraction of the total IND→PR-1 improvement that claim broadcasting alone does not deliver.

#### IV. EXPERIMENTAL DESIGN

All worlds are generated from seeds starting at 1000, disjoint from the curated demonstration seeds of the original release. Each world is run once under every protocol, so all comparisons are paired. Random building placement occasionally walls a victim or the hospital off from both depots; no protocol can rescue such a victim, so these worlds (19 of 1,019 at  $W = 15$ ; 7–8 of 157–158 at  $W = 20$ ) are excluded deterministically before any run and replaced by the next valid seed. Every retained run rescued every victim.

**E1 – Generalization:**  $N = 2$ ,  $M = 6$ ,  $W = 15$ , 1,000 worlds,  $b \in \{0, 0.1\}$ , all six protocols (12,000 runs). This is the original RESQ-MAS setting on random rather than curated worlds. **E2 – Scaling:**  $N \in \{1, 2, 3, 4, 6, 8\}$ ,  $M \in \{10, 20, 30\}$ ,  $W = 20$ , 150 worlds per cell (13,950 runs). **E3 – Independent loss:**  $N = 4$ ,  $M = 20$ ,  $p \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1\}$  for CLM, PR-1, PR-K, CBBA (4,800 runs). **E4 – Blockages:**  $N = 4$ ,  $M = 20$ ,  $b \in \{0, 0.1, 0.2, 0.3\}$ , all protocols (3,600 runs). **E5 – Dynamic arrivals:**  $N = 4$ ,  $M = 20$ ,  $A \in \{0, 100, 200, 400\}$  min, all protocols (3,600 runs). **E6 – Bursty loss:**  $N = 4$ ,  $M = 20$ ,  $p \in \{0.1, 0.3, 0.5\}$ ,  $L_b \in \{1, 5, 20\}$  ticks, CLM, PR-1, PR-K, CBBA (5,400 runs).

Paired differences are tested with the two-sided Wilcoxon signed-rank test (SciPy [16]), with Holm’s step-down correction applied within each experiment family (36 tests in E1, 60 in E2, 48 in E5, 27 in E6). Effect sizes are matched-pairs rank-biserial correlations  $r_{rb}$ , and 95% confidence intervals for mean paired differences use 2,000 bootstrap resamples. The complete study (43,350 runs) takes about 7.5 minutes on four CPU cores.

#### V. RESULTS

##### A. E1: The Original Two-Agent Setting on Random Worlds

Table II and Fig. 2 summarize E1. On 1,000 random worlds, PR-1 reduces completion time from 99.3 to 70.5 minutes (29.0%), mean waiting time by 25.9%, and severity-weighted waiting time by 23.0%, and it eliminates duplicate commitments (4.07 per run to 0) and wasted effort (69.9 agent-minutes to 0). The original release reported a 20.9% completion-time gain on 20 curated runs; the curated benchmark therefore *understated* the effect rather than inflating it. Critical victims, the ones triage protects first under every protocol, still wait 10% less under negotiation (25.2 to 22.7 min), because agents are no longer tied up on duplicates when a critical victim becomes the best candidate.

Table III decomposes the gain. Claim broadcasting alone recovers 11.2 minutes; one-round negotiation adds a further 17.6. In this high-contention setting (three victims per agent), negotiation contributes 61% of the total. Beyond PR-1, returns vanish: PR-K adds 0.4 minutes, HUN a further 0.2, and CBBA lands exactly on PR-K (70.1 min) while transmitting 7.5× more data. These last differences are statistically detectable

TABLE II  
E1: TWO AGENTS, SIX VICTIMS, 1,000 RANDOM WORLDS (MEANS,  $b = 0$ )

| $b$ | Prot. | $T$  | Wait | $\bar{w}_s$ | Crit. | Dup. | Waste | Payl. |
|-----|-------|------|------|-------------|-------|------|-------|-------|
| 0%  | IND   | 99.3 | 54.4 | 45.8        | 25.2  | 4.07 | 69.9  | 0.0   |
| 0%  | CLM   | 88.1 | 50.5 | 43.3        | 25.2  | 2.08 | 43.2  | 8.1   |
| 0%  | PR-1  | 70.5 | 40.3 | 35.2        | 22.7  | 0.00 | 0.0   | 20.4  |
| 0%  | PR-k  | 70.1 | 40.0 | 34.9        | 22.6  | 0.00 | 0.0   | 21.2  |
| 0%  | HUN   | 69.9 | 39.9 | 34.9        | 22.7  | 0.00 | 0.0   | 57.2  |
| 0%  | CBBA  | 70.1 | 40.0 | 35.0        | 22.6  | 0.00 | 0.0   | 158.4 |

All times in simulated minutes.  $\bar{w}_s$ : severity-weighted wait; Crit.: mean wait of critical victims; Dup.: duplicate commitments per run; Waste: agent-minutes on abandoned targets; Payl.: coordination scalars transmitted per run. With 10% blockage no mean changes by more than 1.4 min.

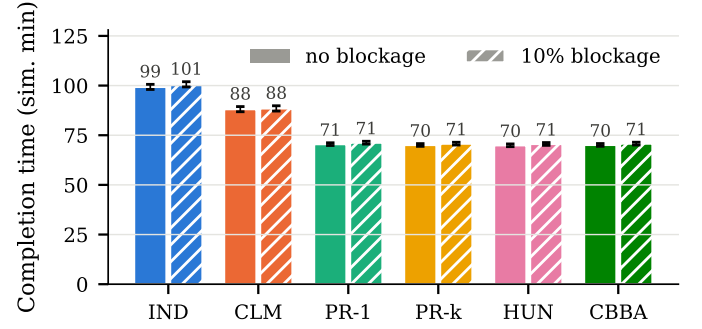


Fig. 2. E1 completion time per protocol with 95% confidence intervals (1,000 paired worlds). Solid bars: no blockage; hatched bars: 10% of road cells blocked mid-run. The ladder improves monotonically, with nearly all of the gain realized by PR-1.

TABLE III  
E1 PAIRED COMPARISONS OF COMPLETION TIME ( $b = 0$ )

| Step      | $\Delta T$ [95% CI] | %    | W/T/L      | $r_{rb}$ | $p_{Holm}$ | % $\bar{w}_s$ |
|-----------|---------------------|------|------------|----------|------------|---------------|
| IND→CLM   | 11.2 [10.3, 12.1]   | 11.3 | 581/385/34 | 0.96     | $<10^{-4}$ | 5.3           |
| CLM→PR-1  | 17.6 [16.5, 18.7]   | 20.0 | 741/227/32 | 0.99     | $<10^{-4}$ | 18.7          |
| IND→PR-1  | 28.8 [27.8, 29.9]   | 29.0 | 972/4/24   | 1.00     | $<10^{-4}$ | 23.0          |
| PR-1→PR-k | 0.4 [0.4, 0.5]      | 0.6  | 436/561/3  | 0.97     | $<10^{-4}$ | 0.8           |
| PR-1→HUN  | 0.6 [0.4, 0.7]      | 0.8  | 452/518/30 | 0.79     | $<10^{-4}$ | 1.1           |
| PR-k→HUN  | 0.2 [0.0, 0.3]      | 0.2  | 63/910/27  | 0.31     | 0.019      | 0.3           |

$\Delta T$ : mean paired reduction (min). W/T/L: worlds where the second protocol is faster / tied / slower. % $\bar{w}_s$ : reduction in severity-weighted wait.

with 1,000 pairs, but 91% of worlds are exact ties between PR-K and HUN and the effect size is small ( $r_{rb} = 0.31$ ). Adding 10% blockages changes no conclusion.

##### B. E2: Uncoordinated Teams Saturate

Fig. 3 and Table IV show the central result. A single agent needs 293, 580, and 870 minutes for 10, 20, and 30 victims. With negotiation, completion time tracks the ideal  $T_1/N$  curve closely: eight agents are 5.0–6.7× faster than one (63–84% parallel efficiency). Without coordination, eight agents are only 1.7–2.5× faster, and doubling the team from four to eight buys 3–12%. The mechanism is visible in Fig. 4(a): independent agents herd onto the same high-priority victims, and wasted effort grows from 241 agent-minutes at  $N = 2$  to 1,662 at

$N = 8$  ( $M = 20$ ). Consequently the empirical price of anarchy rises almost linearly with team size, from 1.26–1.34 at  $N = 2$  to 2.80–3.06 at  $N = 8$ , and PR-1 beats IND on 2,249 of the 2,250 paired worlds with  $N \geq 2$ , the remaining one being a tie (Holm-corrected  $p < 10^{-20}$  in every cell). The starkest comparison is across team sizes: on the same 150 worlds, three negotiating agents finish faster than eight independent ones for every workload ( $M = 10$ : 111 vs. 172 min;  $M = 20$ : 203 vs. 274;  $M = 30$ : 299 vs. 353), winning 129–147 of 150 paired worlds (Wilcoxon  $p < 10^{-21}$ ). Four negotiating agents beat eight independent ones in 449 of 450 worlds and are 35–47% faster. For a dispatcher the practical reading is blunt: buying more vehicles without a coordination layer delivers a fraction of the capacity paid for.

### C. E2: What Negotiation Adds, and When

The negotiation share  $\eta$  in Table IV is not a constant. It rises with the ratio of agents to victims: 4–11% at  $N = 2$ , 13–34% at  $N = 4$ , and 23–50% at  $N = 8$ , reaching 61% in the E1 setting ( $M/N = 3$ ). When victims vastly outnumber agents, agents rarely become idle in the same tick with the same favourite, so public claims prevent almost all duplication and CLM is nearly as good as PR-1. As contention rises, simultaneous choices collide, claims arrive too late, and the proposal round becomes the decisive ingredient. The design rule is therefore conditional: claim broadcasting is sufficient for sparse, large-area operations, while dense incidents with many units need explicit negotiation.

### D. E2: Decentralized Negotiation versus Centralized Optimization

PR-K is statistically indistinguishable from the centralized Hungarian matcher in all 15 ( $N, M$ ) cells after correction, with mean gaps between  $-0.2\%$  and  $+1.2\%$ . The simpler PR-1 is within 3.7% everywhere, but its gap grows with  $N$  and is significant in every cell with  $N \geq 4$  (and in one of the three  $N = 3$  cells): each conflict loser idles for one tick, and with more agents there are more losers per tick. The iterative retry of PR-K removes exactly this cost. The two negotiation protocols also move far less data than HUN, whose agents must report a utility for every candidate victim: at  $M = 30$ , PR-1 transmits 91–152 scalars per mission against roughly 1,000 for HUN. Because HUN optimizes only the current epoch, the decentralized protocols occasionally beat it (negative gaps at  $N = 2$ ).

CBBA also matches HUN: no gap is significant in any cell (largest  $+3.1\%$  at  $N = 8$ ,  $M = 10$ ,  $p_{\text{Holm}} = 0.052$ ). Its bundle look-ahead, however, is inert here: on 39 pilot worlds ( $N = 4$ ,  $M = 20$ ) bundle lengths  $L \in \{1, 2, 3, 5\}$  gave identical makespans, because a busy agent’s future pickup rarely outbids an idle agent’s immediate one, so the converged bundle heads coincide with the sequential greedy auction that PR-K runs. What CBBA adds is traffic: it rebroadcasts full bundles every consensus round (129 rounds per mission at  $N = 4$ ,  $M = 20$ ), transmitting 369–8,321 scalars per mission, 5–56 $\times$  more than PR-1 and 1.9–7.8 $\times$  more than even the centralized matcher.

Fig. 4(b) shows a side effect worth noting. Coordinated teams accumulate *more* idle agent-minutes than independent ones (230 vs. 92 at  $N = 8$ ,  $M = 20$ ), because once every remaining victim is claimed, the surplus agents correctly wait instead of duplicating. That idle time is uncommitted capacity a dispatcher can redeploy, for example to reconnaissance or to a second incident, whereas the independent team’s wasted minutes cannot be recovered.

### E. E3: Robustness to Message Loss

Fig. 5 shows that no protocol collapses under broadcast loss. Completion time rises smoothly from the loss-free value toward the IND level, which all protocols reach exactly at  $p = 1$ , as the model requires. Negotiation retains roughly a  $(1 - p)$  fraction of its gain over IND: 92% at  $p = 0.1$ , 73% at  $p = 0.3$ , and 53% at  $p = 0.5$  for PR-1. With 10% of messages lost, PR-1 is still 42% faster than independent agents. The margin of negotiation over claim-only shrinks with loss (25.2 minutes at  $p = 0$ , 14.7 at  $p = 0.5$ ), because a lost proposal becomes a unilateral commitment. PR-K gains relative to PR-1 as loss grows (8.2 minutes at  $p = 0.7$ ), since its within-tick retries give a losing agent another chance to be heard. CBBA is the most robust protocol under independent loss: at  $p = 0.5$  it still retains 80% of the loss-free gain over IND (184 vs. 217 min for PR-K, which retains 56%), because each agent rebroadcasts its bundle in every consensus round and a receiver needs to hear only one of those copies. Duplicate commitments grow almost linearly in  $p$  (Fig. 5b), so the duplicate rate is a usable online indicator of link quality.

### F. E4: Robustness to Road Blockages

Dynamic blockages of up to 30% of road cells lengthen missions by at most 3.2% under any protocol (PR-1: 158.7 to 163.8 min; IND: 293.1 to 299.5) and leave the IND $\rightarrow$ PR-1 gain essentially constant at 45–46%. Blockages affect routing, which every protocol handles with the same BFS replanning, but not the allocation question that separates them.

### G. E5: Victims Arriving over Time

Fig. 6 and Table V relax the assumption that every victim is known at  $t = 0$ . Coordination matters *more*, not less, when victims trickle in: PR-1 cuts mean victim wait by 44% relative to IND in the static case but by 72–79% for arrival windows of 100–400 minutes, and cuts severity-weighted wait by 78% at  $A = 200$  (143.5 to 31.3 min). All 150 paired worlds favour PR-1 at every window (Holm-corrected  $p < 10^{-23}$ ). The reason is visible in the claim-only column. When a new victim appears, every idle agent sees it in the same tick and it is often everyone’s best candidate, so same-tick collisions become the common case: CLM duplicates rise from 3.7 to 14.0 per run and its wait is 2.4–3.1 $\times$  that of PR-1. The negotiation share  $\eta$  (computed on wait) jumps from 30% to 55–62%. Here, for the first time, CBBA’s look-ahead earns something: letting a unit that is about to drop off at the hospital reserve a fresh arrival lowers wait by a further 0.3–0.9 min relative to PR-K (1–2%,  $p_{\text{Holm}} < 10^{-5}$  for  $A \geq 100$ ), a small but consistent effect.

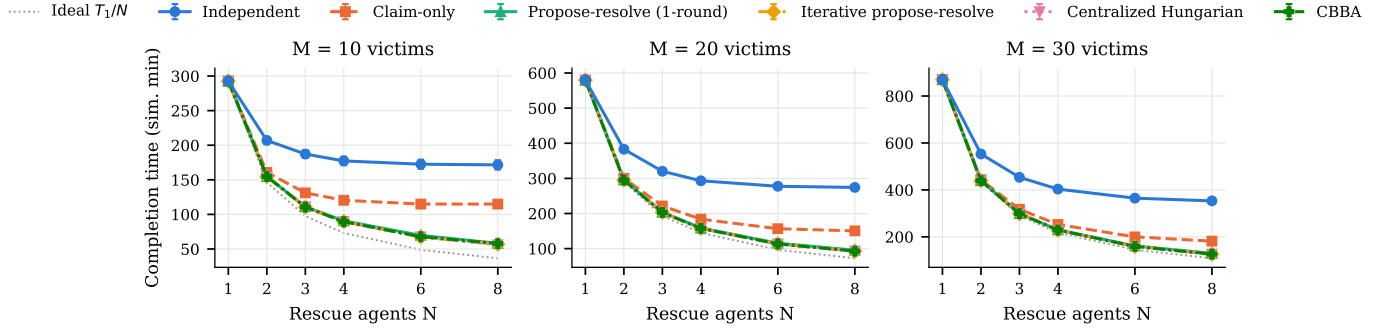


Fig. 3. E2: completion time versus team size  $N$  for three workloads (150 paired worlds per point, 95% CIs). The dotted grey curve is ideal linear speedup from the single-agent time. Independent agents (blue) flatten after  $N \approx 4$ ; claim broadcasting (orange) recovers part of the gap; the four negotiating protocols overlap and stay close to the ideal curve.

TABLE IV  
E2: SCALING, OPTIMALITY GAP, NEGOTIATION SHARE, AND COMMUNICATION VOLUME

| $M$ | $N$ | Completion time $T$ (min) |     |      |     |      |       | Gap to HUN (%) |      |      |            | Speedup $S(N)$ |      | Payload (scalars/run) |      |      |
|-----|-----|---------------------------|-----|------|-----|------|-------|----------------|------|------|------------|----------------|------|-----------------------|------|------|
|     |     | IND                       | CLM | PR-1 | HUN | PoA  |       | PR-1           | PR-K | CBBA | $\eta$ (%) | IND            | PR-1 | PR-1                  | HUN  | CBBA |
| 10  | 2   | 207                       | 161 | 155  | 155 | 1.34 | -0.0  | -0.2           | -0.1 | 11   | 1.41       | 1.89           | 31   | 132                   | 455  |      |
| 10  | 4   | 177                       | 120 | 91   | 90  | 1.98 | +1.2* | -0.0           | -0.2 | 34   | 1.65       | 3.23           | 41   | 145                   | 541  |      |
| 10  | 8   | 172                       | 115 | 58   | 56  | 3.06 | +3.7* | +1.2           | +3.1 | 50   | 1.71       | 5.03           | 68   | 193                   | 369  |      |
| 20  | 2   | 383                       | 300 | 295  | 296 | 1.30 | -0.1  | -0.1           | -0.1 | 6    | 1.51       | 1.96           | 61   | 462                   | 1159 |      |
| 20  | 4   | 293                       | 184 | 159  | 158 | 1.86 | +0.7* | -0.0           | -0.0 | 19   | 1.98       | 3.65           | 72   | 475                   | 2876 |      |
| 20  | 8   | 274                       | 150 | 95   | 93  | 2.96 | +2.9* | +0.1           | -0.1 | 31   | 2.11       | 6.07           | 118  | 532                   | 2486 |      |
| 30  | 2   | 553                       | 443 | 439  | 440 | 1.26 | -0.1  | -0.1           | -0.1 | 4    | 1.57       | 1.98           | 91   | 993                   | 1860 |      |
| 30  | 4   | 403                       | 253 | 230  | 229 | 1.76 | +0.6* | +0.1           | +0.2 | 13   | 2.16       | 3.78           | 103  | 1006                  | 5758 |      |
| 30  | 8   | 353                       | 182 | 130  | 126 | 2.80 | +2.5* | +0.0           | +0.0 | 23   | 2.46       | 6.71           | 152  | 1065                  | 8321 |      |

PoA =  $T_{\text{IND}}/T_{\text{HUN}}$ . Gap:  $(T_X - T_{\text{HUN}})/T_{\text{HUN}}$ ; \* marks Holm-corrected Wilcoxon  $p < 0.05$  (60 tests); negative values mean the protocol beat the per-epoch optimum.  $\eta$ : negotiation share, Eq. (3). Payload: scalars transmitted per run. 150 paired worlds per row.

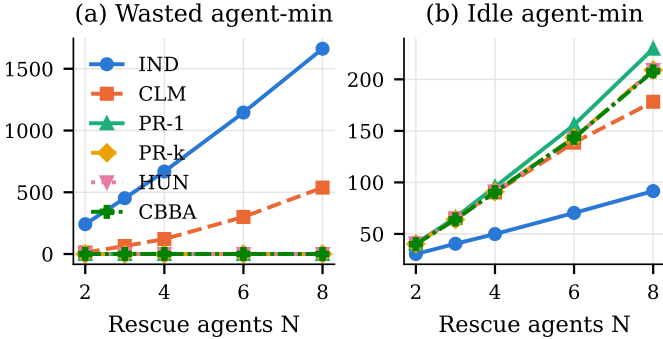


Fig. 4. E2 ( $M = 20$ ): (a) agent-minutes spent on targets later abandoned; (b) agent-minutes idle while victims remained. Negotiation converts wasted travel into idle, redeployable capacity.

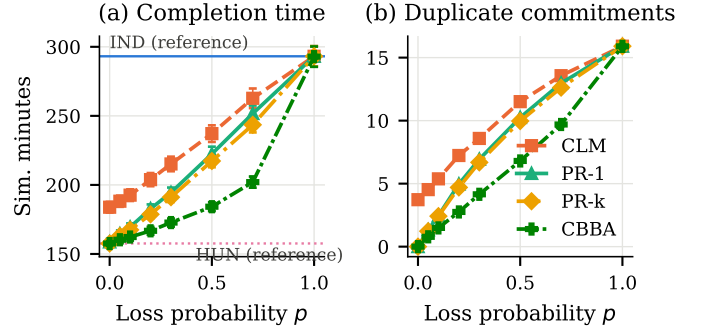


Fig. 5. E3 ( $N = 4$ ,  $M = 20$ , 150 paired worlds per point): (a) completion time and (b) duplicate commitments versus independent broadcast-loss probability. Horizontal lines give the loss-free IND and HUN references. All protocols converge to IND at  $p = 1$ .

#### H. E6: Bursty Outages

Table VI replaces independent loss with Gilbert–Elliott outages of the same average rate. CBBA’s advantage largely disappears. At  $p = 0.5$  its lead over PR-K shrinks from 33 minutes (15%) under independent loss to 6–10 minutes (3–4.5%) under bursts, and at  $p = 0.3$  from 18 minutes to 2–7 (not significant for  $L_b = 1$ ). The outage length barely matters (at  $p = 0.5$ , CBBA takes 206, 203, and 203 min for

$L_b = 1, 5, 20$ ); what matters is that losses are correlated *within* a decision tick. Repeated consensus rounds defeat independent drops because a receiver needs only one copy to get through, but an agent in outage loses every copy it sends that tick. The one-shot protocols are unaffected or even slightly helped by correlation (PR-1: 222 min under independent loss vs. 212–223 under bursts at  $p = 0.5$ ), and negotiation keeps a 7–14% lead over claim-only in every condition ( $p_{\text{Holm}} < 10^{-5}$ ). The

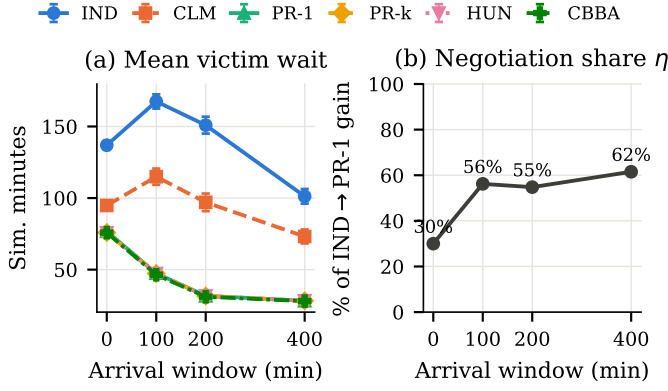


Fig. 6. E5 ( $N = 4$ ,  $M = 20$ , 150 paired worlds per point): (a) mean victim wait, counted from appearance, versus arrival window; (b) negotiation share  $\eta$  of the IND  $\rightarrow$  PR-1 gain in wait. Streaming arrivals make simultaneous choices the norm, so claims alone stop being enough.

TABLE V  
E5: MEAN VICTIM WAIT (MIN) UNDER DYNAMIC ARRIVALS ( $N = 4$ ,  $M = 20$ )

| A   | IND   | CLM   | PR-1 | PR-K | HUN  | CBBA | Red. (%) | Dup <sub>CLM</sub> |
|-----|-------|-------|------|------|------|------|----------|--------------------|
| 0   | 136.9 | 94.8  | 76.7 | 75.9 | 75.7 | 75.9 | 44       | 3.7                |
| 100 | 167.5 | 114.9 | 47.5 | 47.3 | 47.3 | 46.4 | 72       | 7.2                |
| 200 | 150.9 | 97.0  | 31.7 | 31.6 | 31.6 | 31.1 | 79       | 9.3                |
| 400 | 101.2 | 73.1  | 28.3 | 28.3 | 28.2 | 28.0 | 72       | 14.0               |

A: arrival window (min). Red.: IND  $\rightarrow$  PR-1 reduction in mean wait. Dup<sub>CLM</sub>: duplicate commitments per run under claim-only.

TABLE VI  
E6: COMPLETION TIME (MIN) UNDER INDEPENDENT (BERN.) AND BURSTY LOSS ( $N = 4$ ,  $M = 20$ )

| $p$ | $L_b$ | CLM   | PR-1  | PR-K  | CBBA         |
|-----|-------|-------|-------|-------|--------------|
| 0.1 | Bern. | 192.6 | 169.4 | 167.5 | <b>162.3</b> |
|     | 1     | 194.6 | 167.7 | 166.2 | <b>165.5</b> |
|     | 5     | 193.7 | 167.8 | 166.2 | <b>165.1</b> |
|     | 20    | 195.0 | 168.3 | 166.8 | <b>164.8</b> |
| 0.3 | Bern. | 215.2 | 194.5 | 191.1 | <b>172.7</b> |
|     | 1     | 213.7 | 187.1 | 182.3 | <b>180.4</b> |
|     | 5     | 217.8 | 187.1 | 186.5 | <b>182.3</b> |
|     | 20    | 215.5 | 189.3 | 189.0 | <b>181.8</b> |
| 0.5 | Bern. | 237.1 | 222.4 | 217.3 | <b>184.2</b> |
|     | 1     | 240.1 | 223.1 | 215.8 | <b>206.1</b> |
|     | 5     | 240.0 | 211.9 | 209.0 | <b>202.9</b> |
|     | 20    | 239.2 | 213.4 | 210.0 | <b>203.1</b> |

$L_b$ : mean outage length in ticks; Bern.: independent per-message loss at the same rate (from E3). Bold: best in row. Loss-free references: PR-K 157.6, IND 293.1 min.

practical conclusion is that redundancy through repetition, the mechanism behind CBBA’s robustness, should be credited only against the loss pattern it was tested under.

## VI. DISCUSSION

### A. Implications for Dispatch System Design

Three practical rules follow. *First, coordinate before you scale*: without a coordination layer the marginal value of

each added unit falls quickly, so fleet expansion and coordination infrastructure should be planned together. *Second, match the protocol to contention*: when units are few relative to casualties and all casualties are known, lightweight claim broadcasting (a status message per dispatch) captures most of the benefit; in dense incidents, and whenever casualties are reported over time, add a proposal round. *Third, centralization is not required for near-optimal allocation*: iterative propose–resolve matches a central optimizer with a fraction of the traffic and without a single point of failure, which matters when the operations center itself may be unreachable. *Fourth, pay for redundancy only if losses are independent*: CBBA’s repeated consensus rounds cost 5–56 $\times$  the traffic of one-round negotiation and pay off against random packet drops, but not against the correlated outages typical of damaged infrastructure. These rules apply to ambulance and drone dispatch, warehouse and delivery robot fleets, and any setting where independent agents choose from a shared pool of tasks.

### B. Comparison with Existing Work

Our near-equivalence between iterative negotiation and optimal assignment is consistent with the near-optimality of greedy auctions summarized by Quinton *et al.* [3], and our graceful-degradation result under message loss agrees qualitatively with Otte *et al.* [12] and with the 2026 benchmark of Lott and Honary [14]. Those studies compare allocators within the auction and consensus family; we include CBBA from that family and find it equivalent in allocation quality to much cheaper negotiation, with its robustness advantage confined to independent loss, which extends the degraded-channel picture of [14]. What they do not provide, and this one does, is a decomposition of the gain from no communication up to full centralization on a single codebase, which shows that the value of explicit negotiation depends on contention rather than being universal, and that without coordination, adding agents stops paying off.

### C. Limitations

- The world is a synthetic grid with uniform travel cost, one hospital, and one kit per victim. Arrivals are uniform in time; clustered aftershock-style arrivals, heterogeneous vehicles, and capacities above one are not modelled.
- The duplicated-effort penalty depends on simulator rules, in particular the 40-tick kit timeout and the “already allocated” kit denial. Different rules would change the absolute size of the IND penalty, although not the fact that duplicates waste effort.
- HUN is a per-epoch optimum, not a full-horizon optimum; the optimality gaps are relative to that reference. Our CBBA uses synchronous rounds with full connectivity and replans every tick; asynchronous and multi-hop variants may behave differently under loss.
- Loss is modelled as independent or bursty broadcast erasure on each sender’s uplink; network partitions, receiver-specific loss, and message delay are not modelled.

- Times are simulated minutes produced by the model, not field measurements, and priority weights ( $\lambda, \beta$ ) were fixed rather than tuned.

#### D. Future Work

- Replace the grid with real road networks (e.g., OpenStreetMap extracts) and travel times, and calibrate arrivals against historical emergency-call data.
- Add learned bigraph matching [11], multi-victim capacity, and partitions and delay [13].
- Use the duplicate rate observed online to switch adaptively between claim-only and negotiation modes, and between one-shot and repeated-consensus modes as the loss pattern changes.

### VII. CONCLUSION

We asked how much communication a rescue team needs, and answered it with a six-protocol communication ladder evaluated on 43,350 paired simulations. Coordination is worth 29% of mission time for two agents and up to 66% for eight; without it, teams saturate at  $1.7\text{--}2.5\times$  single-agent speed regardless of size. The value splits between broadcasting claims and negotiating simultaneous choices in a ratio set by contention, and it tilts decisively toward negotiation when victims arrive over time. A fully decentralized iterative negotiation matches both a centralized optimal matcher and CBBA while sending far less data; CBBA's extra traffic buys robustness to random packet loss but little against realistic bursty outages. For practitioners the message is simple: three coordinated units outperform eight uncoordinated ones, so the first few bytes of coordination are worth more than the next five vehicles.

### REPRODUCIBILITY

Code, fixed seeds, all raw per-run CSV files, and the scripts that generate every table and figure are in the RESQ-MAS repository (`experiments/study.py`, `experiments/paper_figures.py`). The 85-test suite covers every protocol, both loss models, dynamic arrivals, and the optimal matcher.

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